

# UNPACKING THE SCIENCE AND PRACTICAL USE BY JEFF ROTHSCCHILD PhD

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Knowledgeiswatt

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## Disclaimer

The information provided in this article is for educational and informational purposes only and should not be considered medical advice. Always consult with a qualified healthcare professional before starting any new supplement or making changes to your diet or exercise routine. Sodium bicarbonate supplementation may not be suitable for everyone, and individual needs and risks vary. The author and publisher are not responsible for any adverse effects resulting from the use or reliance on the information provided in this post.



Jeff giving one of his presentations at a scientific congress.

Sodium bicarbonate, widely recognized as baking soda, is not just a pantry staple; **it's also a potent ergogenic aid in the world of cycling. It offers athletes a pathway to improved performance, particularly in high-intensity efforts where a drop in pH (not lactate buildup) can derail energy production and muscle function.** This article dives into the science, application, and practical considerations for using sodium bicarbonate to enhance cycling performance.

## 1. What is Sodium Bicarbonate?

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Chemically, sodium bicarbonate ( $\text{NaHCO}_3$ ) is a compound with strong alkalizing properties (raises pH). **During short bursts of intense activity, the body relies heavily on anaerobic energy systems** (glycolysis and phosphocreatine breakdown) **to generate ATP** (i.e., energy). **This leads to lactate and hydrogen ion ( $\text{H}^+$ ) accumulation, and it is this buildup of  $\text{H}^+$  that causes muscle acidity and is a key factor in fatigue** (Grgic et al., 2021). The severity of these metabolic changes depends on exercise duration and intensity—short maximal efforts

deplete phosphocreatine, while sustained high-intensity efforts (~30 sec to ~5 min) result in significant H<sup>+</sup> buildup. Therefore, exercise of this duration is more likely to benefit from sodium bicarbonate supplementation in comparison to exercise that is too short or too long to lead to muscle acidosis.

**Sodium bicarbonate combats these effects by increasing extracellular pH and bicarbonate levels, creating a more alkaline environment. This enhances H<sup>+</sup> (and lactate) removal from muscles via monocarboxylate transporters (MCT1 and 4), reducing intramuscular acidosis. Improved acid-base balance delays fatigue and sustains higher glycolytic activity and ATP (energy) resynthesis rates during exercise** (Hollidge-Horvat et al., 2000). Beyond buffering acidity, sodium bicarbonate also preserves muscle function through its impact on ion gradients. It stabilizes key electrolytes like potassium (K<sup>+</sup>), sodium (Na<sup>+</sup>), and calcium (Ca<sup>2+</sup>), which are critical for muscle excitability and contraction.

## 2. Does Sodium Bicarbonate Boost Race Performance?

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While it may not help all people in every situation, **there is overwhelming evidence to support its use as an ergogenic aid for various race scenarios.**

### 1. Short to Medium Duration High-Intensity Events (from ~30 seconds ~to ~12 minutes):

Sodium bicarbonate supplementation has consistently shown improvements in performance metrics such as power output and time-to-exhaustion, particularly when individualized dosing strategies were employed to align supplementation timing with peak blood bicarbonate levels (see more below) (Grgic et al., 2021; Hadzic et al., 2019; Lopes-Silva & Correia-Oliveira, 2023).

### 2. Repeated Bout Scenarios: Events requiring **multiple high-intensity efforts** with short recovery periods, such as track cycling, also benefit from sodium bicarbonate. Studies have reported improved recovery between efforts and enhanced performance in later stages (Grgic et al., 2021; Heibel et al., 2018).

### 3. Road Races: Despite their long duration, sodium bicarbonate may provide benefit during road races, **which are often punctuated by short bursts at high intensity**. A study in trained cyclists found a 3% improvement in power output during a 90-s all-out sprint at the end of a 3-h simulated road race (Dalle et al., 2021).

Generally, performance gains of 2–3% have been reported, which can make a critical difference in competitive cycling. However, individual responses vary, and the benefits depend on factors such as exercise duration, athlete conditioning, and dosing strategy (see more below) (Hadzic et al., 2019; Heibel et al., 2018).

## 3. Does Sodium Bicarbonate Boost Training Adaptations?

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Beyond acute performance enhancement, there's growing interest in the potential for sodium bicarbonate to amplify training adaptations ([as reported in a previous KisW Research Note](#)). Although there is evidence that bicarb supplementation around training sessions may actually improve some adaptations (Rothschild & Bishop, 2020), the data are limited. Anecdotally, it has also been noted that because bicarb can let you go harder and “dig deeper” on hard workouts, it could make it too hard to recover from workouts particularly if used over consecutive days (van der Poel, 2023).

At this stage, my personal suggestion is to use it in training only enough to test out the best timing and dosage strategies to adopt in races.

## 4. How to use Sodium Bicarbonate in races?

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Effective use of sodium bicarbonate depends on dosing and timing strategies:

### 1. Dosage

- **The standard dose is 0.2–0.3 g/kg body weight**, consumed 1.5–2.5 h before the event.
- **Start conservatively**, aiming for the lower end of the range (0.2–0.25 g/kg), **gradually increasing** up to 0.35 g/kg based on tolerance.
- For long-term use, multi-day protocols (~3–4 days leading into the target day) can be effective. A total daily dose of 0.4–0.5 g/kg, divided into smaller doses across the day (e.g., with meals), minimizes side effects while maintaining ergogenic benefits. However, in practice, most athletes prefer to use it acutely around competitions.
- Its use during long road races has also been considered, with one study providing half of a dose (0.15 g/kg) before and half of a dose during exercise (Dalle et al., 2021). Some world tour teams have also been reported to be trying this approach. Although there is far less research on peri-exercise bicarbonate ingestion, it may be worth trying.

## 2. Timing

- Peak blood bicarbonate levels occur approximately 60–150 minutes after ingestion, with considerable variation between individuals.
- Individualized strategies, where timing is adjusted to match peak blood bicarbonate levels, have shown superior results. Unfortunately, repeated blood tests are needed to find this out, something only accessible to a very small number of people. It could be worth enquiring with your local University, who may have an exercise physiology lab capable of measuring this.
- **If you don't have the ability to measure your blood bicarbonate levels, aim to consume the full serving 120 min before exercise begins. It is better to err on the earlier side, because when it peaks it will stay elevated for some time.**

## 5. Managing Sodium Bicarbonate Side Effects

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While sodium bicarbonate is effective, its use can be hampered by gastrointestinal (GI) side effects, including nausea, bloating, and diarrhea. These are often dose-dependent and more likely with rapid ingestion of large quantities. Strategies to mitigate these issues include:

- Splitting the dose over multiple smaller intakes. For example, if the race is at 10 am, consume equal doses at 7:30, 7:45, and 8:00 am.
- Using enteric-coated capsules to protect the stomach lining.
- Combine bicarbonate with a carbohydrate-rich meal. This can include a banana, gel, sports drink, or something larger like a bowl of oats or rice.
- Using the Maurten bicarb system, which seems to reduce GI symptoms by embedding bicarbonate mini-tablets within a carbohydrate hydrogel, reducing the interaction with stomach acid (Gough & Sparks, 2024a, 2024b).
- **When to try it:** the first time to try it would be a training session that isn't critical, but includes some high intensity efforts. This allows an athlete to get a feel for the supplement, and how it affects their gut, in a low risk setting while also allowing the potential to observe some performance improvements. It should be trialed at least 3–4 times prior to using it in a race setting.

## 6. What is the Best Sodium Bicarbonate Form?

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Sodium bicarbonate is available in various forms, each with advantages and limitations. In addition, depending on what country you are in there may be different options available.

- **Capsules**

Capsules can reduce GI distress and are easy to consume. **Enteric-coated options are even better if you can get them.**

- **Solutions**

The cheapest option is to mix baking soda into water. **This allows for quick absorption but can increase the risk of stomach upset.**

- **Maurten bicarb system**

The most expensive option, but **early research supports its claims of being an effective ergogenic aid with minimal GI effects.**

- **Topical bicarbonate:** There is a product that claims to offer the benefit of bicarbonate via transdermal (skin) delivery. However, early research does not support the claims being made by the manufacturer (Gurton et al., 2024; Gurton et al., 2023).

## 7. Does Combining Sodium Bicarbonate with Other Supplements Enhance Effects?

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Sodium bicarbonate's effectiveness may be further enhanced when combined with other ergogenic aids, however, research findings are somewhat mixed.

- **Beta-Alanine:** Beta-alanine is an intracellular buffer and sodium bicarbonate is an extracellular buffer. It would be reasonable to consider their combination to be useful, and although most studies show a small, non-significant effect, a meta-analysis found a small positive additive effect of combined supplementation (Saunders et al., 2017).
- **Caffeine:** It is unclear if there are additive effects, but because they work via different pathways, there is no reason to think you should use one or the other. Most likely the context will determine the benefit; exercise intensity and duration, number of repeated bouts, etc.
- **Creatine:** There are limited data, but there may be a positive additive effect (Barber et al., 2013).
- **Nitrates:** The interaction between nitrates and bicarbonate remains unclear, but it has been suggested there may be potential for a negative effect, because the conversion of nitrate to nitric oxide occurs in an acidic environment (Callahan et al., 2017; Grgic et al., 2021).

## 8. Individual Variability

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One of the challenges in using sodium bicarbonate is the variability in individual responses. Athletes are encouraged to experiment with dosing protocols during training to fine-tune their response before competition.

## 9. Summary

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**Sodium bicarbonate is a proven ergogenic aid for cyclists, with well-documented benefits for high-intensity efforts, repeated sprints, and time-trials. While it holds promise for enhancing training adaptations, its primary strength lies in acute performance enhancement.**

To maximize its benefits, athletes should adopt individualized protocols, balancing effective dosing with strategies to mitigate side effects. As with any supplement, sodium bicarbonate works best as part of a comprehensive training and nutrition plan, ensuring the athlete is fully prepared to perform at their peak.

Thanks for reading! Share with your Team!

**Jeff Rothschild, PhD**

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